

p-SERIES

The series in Example 4 are special cases of a class of series called **p-series** or **hyperharmonic series**. A p-series is an infinite series of the form

$$\sum_{k=1}^{\infty} \frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{k^p} + \cdots$$

where $p > 0$. Examples of p-series are

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} + \cdots \quad p = 1$$

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{k^2} + \cdots \quad p = 2$$

$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{k}} + \cdots \quad p = \frac{1}{2}$$

The following theorem tells when a p-series converges.

9.4.5 THEOREM (Convergence of p-Series)

$$\sum_{k=1}^{\infty} \frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{k^p} + \cdots$$

converges if $p > 1$ and diverges if $0 < p \leq 1$.

The Limit Comparison Test

9.5.4 THEOREM (The Limit Comparison Test) Let $\sum a_k$ and $\sum b_k$ be series with positive terms and suppose that

$$\rho = \lim_{k \rightarrow +\infty} \frac{a_k}{b_k}$$

If ρ is finite and $\rho > 0$, then the series both converge or both diverge.

The cases where $\rho = 0$ or $\rho = +\infty$ are discussed in the exercises (Exercise 56).

To use the limit comparison test we must again first guess at the convergence or divergence of $\sum a_k$ and then find a series $\sum b_k$ that supports our guess. The following example illustrates this principle.

► **Example 2** Use the limit comparison test to determine whether the following series converge or diverge.

$$(a) \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}+1} \quad (b) \sum_{k=1}^{\infty} \frac{1}{2k^2+k} \quad (c) \sum_{k=1}^{\infty} \frac{3k^3-2k^2+4}{k^7-k^3+2}$$

Solution (a). As in Example 1, Principle 9.5.2 suggests that the series is likely to behave like the divergent p-series (1). To prove that the given series diverges, we will apply the limit comparison test with

$$a_k = \frac{1}{\sqrt{k}+1} \quad \text{and} \quad b_k = \frac{1}{\sqrt{k}}$$

We obtain

$$\rho = \lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = \lim_{k \rightarrow +\infty} \frac{\sqrt{k}}{\sqrt{k}+1} = \lim_{k \rightarrow +\infty} \frac{1}{1 + \frac{1}{\sqrt{k}}} = 1$$

Since ρ is finite and positive, it follows from Theorem 9.5.4 that the given series diverges.

Solution (b). As in Example 1, Principle 9.5.3 suggests that the series is likely to behave like the convergent series (2). To prove that the given series converges, we will apply the limit comparison test with

$$a_k = \frac{1}{2k^2 + k} \quad \text{and} \quad b_k = \frac{1}{2k^2}$$

We obtain

$$\rho = \lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = \lim_{k \rightarrow +\infty} \frac{2k^2}{2k^2 + k} = \lim_{k \rightarrow +\infty} \frac{2}{2 + \frac{1}{k}} = 1$$

Since ρ is finite and positive, it follows from Theorem 9.5.4 that the given series converges, which agrees with the conclusion reached in Example 1 using the comparison test.

Solution (c). From Principle 9.5.3, the series is likely to behave like

$$\sum_{k=1}^{\infty} \frac{3k^3}{k^7} = \sum_{k=1}^{\infty} \frac{3}{k^4} \quad (3)$$

which converges since it is a constant times a convergent p -series. Thus, the given series is likely to converge. To prove this, we will apply the limit comparison test to series (3) and the given series. We obtain

$$\rho = \lim_{k \rightarrow +\infty} \frac{\frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}}{\frac{3}{k^4}} = \lim_{k \rightarrow +\infty} \frac{3k^7 - 2k^6 + 4k^4}{3k^7 - 3k^3 + 6} = 1$$

Since ρ is finite and nonzero, it follows from Theorem 9.5.4 that the given series converges, since (3) converges. ◀

Practice Questions

5–10 Use the limit comparison test to determine whether the series converges. ■

5. $\sum_{k=1}^{\infty} \frac{4k^2 - 2k + 6}{8k^7 + k - 8}$

6. $\sum_{k=1}^{\infty} \frac{1}{9k + 6}$

7. $\sum_{k=1}^{\infty} \frac{5}{3k + 1}$

8. $\sum_{k=1}^{\infty} \frac{k(k + 3)}{(k + 1)(k + 2)(k + 5)}$

9. $\sum_{k=1}^{\infty} \frac{1}{\sqrt[3]{8k^2 - 3k}}$

10. $\sum_{k=1}^{\infty} \frac{1}{(2k + 3)^{17}}$